Year 12 Maths - Using graphs

Similar questions - Direct and inverse proportion

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The kinetic energy, E joules, of a moving object of fixed mass is directly proportional to the square of its speed, v m s^{-1}. When $v = 6$, $E = 90$.

(a) Find an equation for E in terms of v. **[2]**

(b) Find the speed when $E = 360$. **[2]**

(c) The speed is increased by 50%. Find the percentage increase in E. **[3]**

2 y is inversely proportional to the cube of x, and $y = 5$ when $x = 2$.

(a) Find an equation connecting y and x. **[2]**

(b) Find the value of x when $y = 0.625$. **[2]**

(c) State what should be plotted on each axis to give a straight-line graph through the origin, and give its gradient. **[2]**

Question 1

(a) $E = kv^2$; $90 = 36k$ so $k = 2.5$ and $E = 2.5v^2$

(b) $360 = 2.5v^2$, so $v^2 = 144$ and $v = 12$ m s^{-1}

(c) New speed $1.5v$ gives $E = 2.5(1.5v)^2 = 2.25 \times 2.5v^2$, so E is multiplied by 2.25: a 125% increase

Question 2

(a) $y = \dfrac{k}{x^3}$; $5 = \dfrac{k}{8}$ so $k = 40$ and $y = \dfrac{40}{x^3}$

(b) $0.625 = \dfrac{40}{x^3}$, so $x^3 = 64$ and $x = 4$

(c) Plot y against $\dfrac{1}{x^3}$; the gradient is 40